

Level	Phase	Cognitive control state	Grounding	occ.
L1 goal-setting	PLAN	goal decomposition; PFC sets the goal hierarchy	[2, 4]	0
L2 execution	RETRIEVE	directed, externally-directed acquisition	[5]	4,298
L2 execution	INSPECT	metacognitive monitoring; no proposition tested	[6]	4,079
L2 execution	EXTRACT	central executive locks on a target value	[7]	1,681
L2 execution	VERIFY	metacognitive control; output is a truth value	[6, 8]	7,931
L2 execution	WRITE	generative production; externalisation mode	[9]	5,824
L3 integration	SYNTHESIZE	multi-source integration into a new representation	[10]	0
L4 interrupt	REPAIR	error monitoring; SAS override after a failure signal	[11]	116
L4 interrupt	HANDOFF	distributed cognitive load transfer (agent change)	[10, 12]	0

Table 1: The nine cognitive phases over four abstraction levels, with their control-state definitions and grounding. The *occ.* column gives the per-phase occurrence count over the 500-trajectory corpus. The empty L3/L4 phases are a finding, not a gap: single-agent coding traces do not integrate across sources (SYNTHESIZE) or transfer control (HANDOFF), and overt planning (PLAN) is latent in reasoning the trace does not externalise.

1 Procedural annotation framework

We annotate each agent trajectory in three layers, following a standoff multi-layer design conformant to the Linguistic Annotation Framework (ISO 24612:2012; [1]).

Layer B.1a: action tokens. Each raw event is normalised, with *no* interpretive rule, into an *action token* carrying four fields, $a = (action_type, tool_family, agent_id, artifact_id)$, and adjacent identical tokens are merged (de-duplication). This is pure data cleaning and is the substrate every downstream layer references.

Layer A: cognitive phases. Each span is assigned, *first*, one of nine cognitive control states grounded in the prefrontal-cortex theory of cognitive control [2]. The phases are organised over four abstraction levels — L1 goal-setting (PLAN); L2 execution (RETRIEVE, INSPECT, EXTRACT, VERIFY, WRITE); L3 integration (SYNTHESIZE); L4 interrupt (REPAIR, HANDOFF). The phase label is an *optional, coarse procedural descriptor* used for interpretability and diagnostic analysis; it is *not* a fixed skill type. HPOP does not take phases as skill labels: it receives the CPA occurrence sequences and learns a reusable library of *local partial-order skills*, and a learned skill may contain CPAs from one phase or several. The phase taxonomy is given in Table 1.

Layer B.1b: canonical procedural actions. Within each phase, the span is refined to a *canonical procedural action* (CPA) — the procedural *function* realising that control state, more abstract than a tool call and portable across repositories. CPAs correspond to the *Action* tier of Activity Theory [3]: goal-directed, consciously executed units, as opposed to the automatised

Operations (raw clicks/commands) at which reusable structure does not emerge. The annotation is thus coarse-to-fine: phase first, CPA within it, each CPA realising exactly one phase.

1.1 Coarse-to-fine generative model

We model annotation generatively as *phase first*, CPA *second*. Each phase $h \in \mathcal{P}$ owns a phase-specific dictionary $\mathcal{A}_h = \{c_{h1}, \dots, c_{hK_h}\}$, and the full action vocabulary is $\mathcal{A} = \bigcup_{h \in \mathcal{P}} \mathcal{A}_h$. For an event span x_i with local context ξ_i (preceding spans, edit/run history, artifact state),

$$h_i \sim p(h_i \mid \xi_i), \quad (\text{phase}) \quad (1)$$

$$c_i \sim p(c_i \mid h_i, \mathcal{A}_{h_i}), \quad (\text{CPA within phase}) \quad (2)$$

$$x_i \sim p(x_i \mid c_i, h_i), \quad (\text{realisation}) \quad (3)$$

so that, over a trajectory,

$$p(H, C, X) = \prod_i p(h_i \mid \xi_i) p(c_i \mid h_i) p(x_i \mid c_i, h_i), \quad (4)$$

with annotation posterior $p(h_i, c_i \mid x_i) \propto p(x_i \mid c_i, h_i) p(c_i \mid h_i) p(h_i \mid \xi_i)$. The *generative* direction is $p(c_i \mid h_i)$ — the phase *guides* CPA construction — not the diagnostic posterior $p(h_i \mid c_i)$; both are well defined but answer different questions.

Phase-specific dictionary complexity. Let $K_h = |\mathcal{A}_h|$. We regularise the induced dictionary with a per-phase complexity penalty $-\lambda_{\text{CPA}} \sum_{h \in \mathcal{P}} K_h$, admitting a candidate CPA into phase h only when it explains spans existing CPAs in that phase cannot,

$$\Delta \log p(x_i \mid c_{\text{new}}, h) > \lambda_{\text{CPA}} + \Delta \text{Complexity}, \quad (5)$$

which prevents synonym explosion within a phase.

Three distinct latent objects. We keep separate symbols so that annotation priors are not conflated with learned structure: $h_i \in \mathcal{P}$ is the *phase* (a coarse annotation prior, fixed in preprocessing); $c_i \in \mathcal{A}_{h_i}$ is the CPA (the observed procedural-action label); $z_\ell \in \{1, \dots, K\}$ is the HPOP *skill* (a local partial order over CPA occurrences, learned downstream). In general $h_i \neq z_\ell$: a learned skill may span several phases. The pipeline is

raw trace $\rightarrow$ phase-guided CPA annotation $\rightarrow$ CPA occurrence sequence $\rightarrow$ HPOP skill discovery.

Why phase-first helps. The same physical command takes different procedural meaning by phase. The command `pytest tests/test_parser.py` is (INSPECT, REPRODUCE_FAILURE) before any edit, (VERIFY, RUN_TEST_TO_VALIDATE_PATCH) after an edit, and (REPAIR, CHECK_REPAIR_SIGNAL) following a failure. Conditioning CPA selection on the phase resolves an ambiguity a phase-free labelling cannot.

2 Evaluating the annotation against no-instruction baselines

A labelling scheme assigns a symbol $\ell \in V$ to each occurrence. A *good* scheme makes the trajectory predictable: the current label should be informative about the next. We therefore measure the sequential structure a scheme induces, relative to a null that destroys order.

2.1 Transition information and the shuffle null

For a scheme, collect the multiset of consecutive label pairs over all trajectories, $\mathcal{P} = \{(\ell_t, \ell_{t+1})\}$, and estimate the *transition mutual information*

$$\hat{I} = \sum_{x,y \in V} \hat{p}(x,y) \log_2 \frac{\hat{p}(x,y)}{\hat{p}(x)\hat{p}(y)}, \quad (6)$$

where $\hat{p}(x,y)$ is the empirical joint over adjacent pairs and $\hat{p}(x), \hat{p}(y)$ the current/next marginals. If the next label is independent of the current one, $\hat{I} = 0$.

Finite samples inflate $\hat{I}$ by an $O(|V|^2/N)$ bias, so a larger vocabulary scores spuriously higher. We remove it with a within-trajectory *shuffle null*: permute each trajectory’s labels uniformly at random (preserving length and unigram composition, destroying order), recompute (6), and average over $B = 150$ permutations to obtain $\bar{I}_0$. Because the null shares the exact marginals and N of the data, $\bar{I}_0$ estimates the finite-sample floor for that scheme. We report the *excess* structure and a size-normalised *efficiency*,

$$\Delta = \hat{I} - \bar{I}_0, \quad \eta = \frac{\Delta}{\log_2 |V|}, \quad (7)$$

where $\log_2 |V|$ is the maximum information a single symbol can carry, making schemes of different vocabulary sizes comparable. By construction a labelling with no ordering structure has $\Delta \approx 0$.

2.2 Baselines

All schemes label the *same* occurrence segmentation; only the symbol differs.

cmdword (no instruction). ℓ is the raw leading command token (`pytest`, `grep`, `view`, ...); no taxonomy. A natural, uncontrolled vocabulary.

actobj (no instruction). $\ell = (\text{action_type}, \text{artifact_role})$, e.g. `run:test`, `edit:source`; still no procedural taxonomy.

v1 cpa (induced). The bottom-up dictionary: surface forms mined from data, then grouped into phases post hoc.

v2 cpa (cognitive). The top-down dictionary of Section 1: nine phases first, CPAs derived as their realisations.

2.3 Results

Table 2 reports the ladder on 500 SWE-rebench/OpenHands trajectories (334 repositories; 23,929 occurrences). A *no-instruction* labelling is the honest floor; both principled dictionaries lie well above it, and the cognitive v2 scheme attains the highest excess and the highest per-symbol efficiency.

Information beyond the surface. To confirm the abstraction is not a relabelling of the physical layer, we compare three *layers* of the same trace by (i) normalised predictability $\text{NP} = \hat{I}(\ell_t; \ell_{t+1}) / \hat{H}(\ell_t)$ and (ii) the information a current label gives about the next physical action, $\text{IG} = \hat{H}(a_{t+1}) - \hat{H}(a_{t+1} \mid \ell_t)$. At nearly equal vocabulary the cognitive phase layer ($|V|=6$, $\text{NP}=0.186$, $\text{IG}=0.253$ b) is about twice as self-predictable as the raw physical layer ($|V|=5$, $\text{NP}=0.097$, $\text{IG}=0.159$ b), and a current phase is *more* predictive of the next physical action than the current physical action itself — the abstraction explains the surface, not vice versa.

scheme	$ V $	$\hat{I}$	$\bar{I}_0$	Δ (bits)	η
CMDWORD (no instruction)	62	0.623	0.041	0.582	0.098
ACTOBJ (no instruction)	24	0.941	0.023	0.918	0.200
v1 CPA (induced)	28	0.968	0.027	0.940	0.196
v2 cpa (cognitive)	29	1.009	0.030	0.979	0.201

Table 2: Sequential structure beyond a shuffled null (Eq. 7). The naive no-instruction labelling sprawls to $|V| = 62$ yet is half as efficient ($\eta = 0.098$); the cognitive dictionary packs the most non-random structure into the fewest reusable symbols.

References

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